Fiscal Year:	FY 2022	**Task Last Updated:**	FY 07/13/2023
PI Name:	Basner, Mathias M.D., Ph.D.		
Project Title:	Cognitive Performance in Relation to Commercial Space Crew Behavior and Spacecraft Environment		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Philadelphia	**State:**	PA
Zip Code:	19104-4209	**Congressional District:**	2
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	TRISH--EXPAND
Start Date:	05/01/2021	**End Date:**	04/30/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	SpaceX		
Flight Assignment:	SpaceX Inspiration4 NOTE: End date changed to 04/30/2022 per TIMS/TRISH (Ed., 9/6/23) NOTE: End date changed to 04/30/2023 per TRISH (Ed., 8/4/22)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-EXP0005		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	EXPAND Project The proposed research will be performed on SpaceX's first commercial crew flight: the Inspiration4 (I4) mission. The 4 person crew is expected to launch in the Dragon spacecraft on a Falcon rocket from Kennedy Space Center in September 2021. After 2-3 days in space, they will perform a soft water landing off the coast of Florida. The primary objective of this study is to collect data on I4 crew behavior and performance before, during, and after the mission, to extract the data from the devices that will be used to collect the data, clean and annotate the data, visualize the data for crew debriefs, and submit the data to a database selected by the Translational Research Institute for Space Health (TRISH). Data collected by other investigators and environmental data collected by Dragon will also be submitted to the database. The long-term goal of this TRISH effort is to generate a quickly growing commercial crew database that can be mined by the space research community.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project demonstrates that commercial crews can collect valuable cognitive and physiological data with relatively little effort. These data will help improve health and safety for future crews.
Task Progress:	The research of this project was part of SpaceX's first commercial crew flight: the Inspiration4 (I4) mission. The 4-person crew launched in the Dragon spacecraft from Kennedy Space Center on 9/15 in September 2021. After almost 3 days in space, they performed a soft water landing off the coast of Florida. The primary objective of this study was to collect data on I4 crew behavior and performance before, during and after the mission, to extract the data from the devices that will be used to collect the data, clean and annotate the data, visualize the data for crew debriefs, and submit the data to a database selected by TRISH. Data collected by other investigators and environmental data collected by Dragon will also be submitted to the database. The long-term goal of this TRISH effort is to generate a quickly growing commercial crew database that can be mined by the space research community. Data collection was almost 100% complete, and the data were submitted to the database hosted by TrialX. Main study outcomes were presented at the 2022 NASA Human Research Program (HRP) Investigators' Workshop (IWS) and at a crew debrief meeting in March 2022 in Houston.
Bibliography Type:	Description: (Last Updated: 06/17/2026)